WATER RESOURCES, WELLS & PUMPS · NOTE PU05

Pumping Power, Drives and the Cost of Pumping

Course topics: IDE-511 T7, IDE-511 T10, IDE8.16 L6, CAE1.3 L15

Heads

A pump works best at the head and speed it was designed for, so determine the operating conditions carefully — studying the heads and pump location can save much power over the years at no extra first cost.

Head	Definition
Static head	Vertical distance between the water surface at the source and at the outlet
Static suction head	Vertical distance from the water level to the pump centre line
Static discharge head	Vertical distance from the pump centre line to the point of free delivery
Friction head	Head needed to overcome the pipeline and fittings (strainer, elbows, bends, reducers, tees, valves) on both suction and discharge sides; varies with flow rate, pipe size, pipe material and interior condition
Velocity head	The height through which water must fall to gain its velocity: h = v²/2g
Total head	Static head + friction head + velocity head

$$\text{Static head} = \text{static suction head} + \text{static discharge head}$$

$$\text{Strainer loss } h_f = K_s \times v^2/2g \ (K_s = 0.95) \quad \text{Foot-valve loss } h_f = K_f \times v^2/2g \ (K_f = 0.80)$$

h_f = head lost (m); v = velocity (m/s); g = 9.81 m/s². Read losses in other valves and fittings from standard charts (Fig. 6.49 of the textbook).

Power

- 1 HP = 76 metre-kilograms per second = 0.746 kilowatt-hours per hour (0.746 kW).
- **Water horsepower (WHP)** is the useful power spent lifting water — what the pump and drive would need at 100 per cent efficiency.

$$\text{WHP} = Q\ (\text{L/s}) \times \text{total head (m)} \ / \ 76 \text{ or } \text{WHP} = Q\ (\text{m}^3/\text{hour}) \times \text{total head (m)} \ / \ 273$$

$$\text{Shaft HP} = \text{WHP} \ / \ \text{pump efficiency}$$

Power to be supplied	Formula
Brake HP — direct drive (drive efficiency 100%)	BHP = WHP / pump efficiency = shaft HP
Brake HP — belt or other indirect drive	BHP = WHP / (pump efficiency × drive efficiency)
Overall efficiency	Pump efficiency × drive efficiency
Input HP to an electric motor	BHP / motor efficiency = WHP / (pump eff. × drive eff. × motor eff.)
kW input to the motor	BHP × 0.746 / motor efficiency

> Q = 15 L/s against a total head of 20 m: WHP = 15 × 20 / 76 = 3.95 hp. With 65% pump efficiency, 90% V-belt drive and 85% motor efficiency: BHP = 3.95 / (0.65 × 0.90) = 6.75 hp; kW input = 6.75 × 0.746 / 0.85 ≈ 5.9 kW.

Drives

Drive	Efficiency and remarks
Direct drive	Most efficient — no transmission loss — but only when the prime mover and pump run at the same speed. Horizontal centrifugal pumps use a flexible coupling (the motor can be removed without disturbing the pump and piping) or a close-coupled unit (always aligned, avoiding misalignment losses). For electric vertical turbine pumps it is the cheapest and most efficient: a vertical motor coupled to the pump head, with a non-reverse coupling for water-lubricated and a self-release coupling for oil-lubricated pumps
Flat belt	Least efficient — 80 to 90 per cent, lost mainly to slippage (also depends on pulley and belt type, pulley size, idlers and twist)
V-belt	More dependable, 90 to 95 per cent, pulleys much closer together; costlier than flat belts
Right-angle gear head	The most dependable and efficient way to drive a turbine pump from an engine — about 95 per cent; a range of gear ratios lets pump and engine both run at their best speeds; the drive shaft has a flexible or universal joint; costly
Tractor PTO	Where the water table is not very deep, to use tractor power. Standard PTO speed is 540 ± 10 rpm, while irrigation pumps run at 1450 rpm or more, so a speed increaser is needed — pump and increaser on a two-wheel trailer

Prime movers

- **Electric motor:** most commonly a 50 cycle, 440 volt, 3-phase squirrel-cage induction motor running at 1450 or 2900 rpm. Where power is available it offers low cost, reliability, high efficiency, compactness, low maintenance and little attention. (Indian three-phase LT supply is now standardised at 415 V.)
- **Engine sets:** a vertical diesel engine (commonly 1500 rpm) for direct-coupled units; vertical or horizontal engines for belt drives. Choose an engine rated about 20 per cent above the calculated BHP so it is not overloaded.

Cost of pumping

Fixed costs

$$\text{Annual interest} = (\text{value of installation} \times \text{interest rate}) / 2 \text{ (interest about 12 per cent, on the average value)}$$

$$\text{Annual depreciation} = (\text{original cost} - \text{salvage value}) / \text{useful life (years)}$$

Item (Table 6.1)	Useful life
Well and casing; pump house	20 years each
Turbine pump — bowl (≈ 50% of cost) / column etc.	16,000 hours or 8 years / 32,000 hours or 16 years
Centrifugal pump	32,000 hours or 16 years
Gear head	30,000 hours or 15 years
V-belt	6,000 hours or 3 years
Flat belt — rubber/fabric / leather	10,000 hours or 5 years / 20,000 hours or 10 years
Electric motor	50,000 hours or 25 years
Diesel engine	28,000 hours or 14 years
Petrol engine — air-cooled / water-cooled	8,000 hours or 4 years / 18,000 hours or 9 years

Operating costs

1. **Electricity:** motor efficiency is 80 to 90 per cent; energy per hour (kWh) = BHP / motor efficiency × 0.746.
2. **Fuel:** use the maker's fuel curve, or the averages below; cost per hour = BHP × fuel per hp-hr × fuel price per litre.
3. **Lubricating oil and grease:** negligible for electric sets; about 4.5 litres per 1000 hp-hrs for petrol and diesel engines.
4. **Pump maintenance and repairs:** centrifugal pump — yearly cost = total cost / estimated life (years); turbine pump — yearly cost = one-half total cost / estimated life (years).
5. **Engine maintenance:** hard to predict — add a nominal amount. **Operator's wages:** at prevailing rates.

Engine (Table 6.2)	Average fuel use
Petrol, air-cooled	0.5 litres/hp-hr
Petrol, water-cooled	0.4 litres/hp-hr
High-speed diesel	0.32 litres/hp-hr

Corrected factor. The source notes print the energy formula with 0.7466. One horsepower is 0.746 kW (more exactly 0.7457), so use kWh per hour = BHP / motor efficiency × 0.746.

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Lessons on pumps (classification, centrifugal and turbine pumps, pumping power). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Water horsepower for Q in L/s and head H in m is

(a) QH/75 (b) QH/76
(c) QH/273 (d) QH/102

Q2. Total head is the sum of

(a) suction and delivery heads only (b) static, friction and velocity heads
(c) velocity and pressure heads (d) static head and drawdown

Q3. Efficiency of a flat-belt drive is about

(a) 50–60% (b) 80–90%
(c) 95–98% (d) 100%

Q4. Standard tractor PTO speed is

(a) 540 ± 10 rpm (b) 1000 rpm
(c) 1450 rpm (d) 2900 rpm

Q5. An engine for a pump should be rated above the calculated BHP by about

(a) 5% (b) 20%
(c) 50% (d) 100%

Q6. Average fuel consumption of a high-speed diesel engine is about

(a) 0.32 L/hp-hr (b) 0.5 L/hp-hr
(c) 1.0 L/hp-hr (d) 3.2 L/hp-hr

Answers

Q1 — (b) QH/76 — *With Q in m³/h the divisor is 273.*
Q2 — (b) static, friction and velocity heads
Q3 — (b) 80–90%
Q4 — (a) 540 ± 10 rpm
Q5 — (b) 20%
Q6 — (a) 0.32 L/hp-hr